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CODE:

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# Credit Score Classification

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# Create sample credit dataset
data = {
    "Age": [25, 35, 45, 30, 50, 28, 40, 55, 32, 48, 60, 27],

    "Annual_Income": [
        30000, 60000, 90000, 45000,
        120000, 35000, 75000, 150000,
        50000, 100000, 180000, 25000
    ],

    "Monthly_Balance": [
        2000, 5000, 8000, 3500,
        10000, 2500, 6000, 12000,
        4000, 9000, 15000, 1500
    ],

    "Num_Credit_Cards": [
        2, 3, 2, 5,
        2, 4, 3, 1,
        4, 2, 1, 6
    ],

    "Num_Bank_Accounts": [
        2, 3, 2, 4,
        1, 3, 2, 1,
        4, 2, 1, 5
    ],

    "Credit_Score": [
        "Poor", "Standard", "Good", "Standard",
        "Good", "Poor", "Good", "Good",
        "Standard", "Good", "Good", "Poor"
    ]
}

# Convert data into a DataFrame
df = pd.DataFrame(data)

# Display dataset
print("CREDIT SCORE DATASET")
print("-" * 50)
print(df)

# Input features
X = df.drop("Credit_Score", axis=1)

# Target variable
y = df["Credit_Score"]

# Split dataset into training and testing data
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X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)

# Create Random Forest model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

# Train the model
model.fit(X_train, y_train)

# Predict the test data
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

# Display results
print("\nMODEL RESULTS")
print("-" * 50)

print("Accuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred, zero_division=0))

# Classify a new customer's credit score
new_customer = pd.DataFrame([
    "Age": 38,
    "Annual_Income": 70000,
    "Monthly_Balance": 5500,
    "Num_Credit_Cards": 3,
    "Num_Bank_Accounts": 2
])

prediction = model.predict(new_customer)

print("\nNEW CUSTOMER DETAILS")
print("-" * 50)
print(new_customer)

print("\nPredicted Credit Score:", prediction[0])

```

OUTPUT :

Classification Report:

	precision	recall	f1-score	support
Good	1.00	1.00	1.00	2
Poor	1.00	1.00	1.00	1
accuracy			1.00	3
macro avg	1.00	1.00	1.00	3
weighted avg	1.00	1.00	1.00	3

NEW CUSTOMER DETAILS

	Age	Annual_Income	Monthly_Balance	Num_Credit_Cards	Num_Bank_Accounts
0	38	70000	5500	3	2

Predicted Credit Score: Good